

Disproof of TLMC Conjecture 00000001168

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Verdict

FALSE. The round metric on $\mathrm{SU}(2) \cong \mathbb{S}^3$ is itself a Zoll metric, and its first nonzero Laplace eigenvalue has multiplicity $4 > 2$; the family of rotated round metrics that the conjecture itself cites violates the claimed bound on every member.

1 The conjecture

Definition: A Zoll metric is one all of whose geodesics are closed. *Conjecture:* The spectral multiplicity upper bound for Zoll metrics on $\mathrm{SU}(2)$ is two; and the counterexample to multiplicity rigidity is the family of rotated round metrics. (Zoll multiplicity rigidity)

2 Counterexample: the round metric on $\mathrm{SU}(2)$

Identify $\mathrm{SU}(2)$ with the unit quaternions, i.e. with the round unit sphere $\mathbb{S}^3 \subset \mathbb{R}^4$. Write g_r for the round (constant curvature 1) metric.

Lemma 1. $(\mathbb{S}^3, g_r)$ is a Zoll metric in the sense of the conjecture's own definition: every geodesic is a great circle $\gamma(t) = \cos(t)u + \sin(t)v$ with $u, v \in \mathbb{S}^3$ orthonormal, closed with the common period 2π .

Proof. The geodesics of the round sphere are exactly the great circles $C = \mathbb{S}^3 \cap P$, $P \subset \mathbb{R}^4$ a two-plane through the origin, and the parametrization above (unit speed) satisfies the geodesic equation $\ddot{\gamma} + \gamma = 0$ identically; since $\cos$ and $\sin$ have common period 2π , $\gamma(t + 2\pi) = \gamma(t)$ for every great circle. The script `reproduce.py` verifies $|\gamma(t)| = 1$, $\ddot{\gamma} + \gamma = 0$ at sample points, and the 2π -periodicity for 200 sampled orthonormal pairs (u, v) . $\square$

Theorem 2. The k -th eigenvalue of the Laplace–Beltrami operator of $(\mathbb{S}^3, g_r)$ is $\lambda_k = k(k + 2)$, $k = 0, 1, 2, \dots$, with multiplicity $(k + 1)^2$. In particular $\lambda_1 = 3$ has multiplicity $4 > 2$. Hence the round metric g_r , transported to $\mathrm{SU}(2)$ along the quaternion identification, is a Zoll metric on $\mathrm{SU}(2)$ whose spectrum violates the claimed upper bound of two. The conjecture is false.

Proof. The eigenfunctions on the round $\mathbb{S}^3 \subset \mathbb{R}^4$ are the restrictions of the homogeneous harmonic polynomials on $\mathbb{R}^4$: the eigenspace for λ_k is the restriction of $\mathcal{H}_k$, the kernel of the flat Laplacian Δ restricted to the space Hom_k of homogeneous polynomials of degree k in 4 variables.

Dimension count. $\dim \mathrm{Hom}_k$ is the number of monomials of degree k in 4 variables,

$$\dim \mathrm{Hom}_k = \binom{k+3}{3} = \frac{(k+1)(k+2)(k+3)}{6}.$$

Since Δ maps Hom_k onto Hom_{k-2} (classical; Δ is surjective because it is self-adjoint for the Fischer inner product and its image is the orthogonal complement of the harmonic space in Hom_{k-2} ; surjectivity is also verified computationally below), and $\dim \text{Hom}_{k-2} = \binom{k+1}{3}$, we get

$$\dim \mathcal{H}_k = \binom{k+3}{3} - \binom{k+1}{3} = (k+1)^2.$$

The case $k = 1$ explicitly. Hom_1 is the space of linear forms $c_1x_1 + c_2x_2 + c_3x_3 + c_4x_4$, all of which are harmonic; so $\lambda_1 = 1 \cdot 3 = 3$ and the eigenspace contains the restrictions of the four coordinate functions. These are linearly independent: the evaluation matrix at the four axis points $e_1, \dots, e_4 \in \mathbb{S}^3$ is the identity ($x_j(e_i) = \delta_{ij}$), so a linear combination vanishing on $\mathbb{S}^3$ has $c_i = 0$ for every i . Hence $\dim \mathcal{H}_1 = 4$, matching $(1+1)^2 = 4$.

Independent computational verification. The script `reproduce.py` assembles the flat Laplacian $\Delta: \text{Hom}_k \rightarrow \text{Hom}_{k-2}$ as an explicit matrix over $\mathbb{Q}$ (entry $\partial^2/\partial x_i^2 x^a = a_i(a_i - 1)x^{a-2e_i}$) and computes its rank by exact rational Gaussian elimination for $k = 0, \dots, 8$. In every case the map is surjective onto Hom_{k-2} and the kernel has dimension exactly $(k+1)^2$:

k	0	1	2	3	4	5	6	7	8
$\lambda_k = k(k+2)$	0	3	8	15	24	35	48	63	80
multiplicity $(k+1)^2$	1	4	9	16	25	36	49	64	81

Every nonzero eigenvalue has multiplicity > 2 ; already λ_1 has multiplicity 4. □

3 The conjecture's own example family refutes its own bound

The conjecture proposes “the family of rotated round metrics” as the counterexample to multiplicity rigidity. But a rotation $Q \in SO(4)$ is an isometry of $(\mathbb{S}^3, g_r)$: it preserves round distances ($Q^T Q = I$), so the pulled-back metric Q^*g_r is *identically* g_r . The “family of rotated round metrics” is a single isometry class carrying a single spectrum, with $\text{mult}(\lambda_k) = (k+1)^2$ on every member — in particular $\text{mult}(\lambda_1) = 4 > 2$ on every member. So the proposed family does not merely fail to support the claimed upper bound of two; it is precisely a set of witnesses against it. The two clauses of the conjecture are jointly inconsistent:

- the bound clause (“multiplicity upper bound is two”) is refuted by g_r and by every rotated round metric, all of which are Zoll metrics on $\text{SU}(2)$;
- the counterexample clause is vacuous as stated, because the proposed counterexample family is one isometry class with no metric-to-metric spectrum variation to exhibit, and it contradicts the bound clause.

A single counterexample suffices to refute the universal bound: the round metric g_r is a Zoll metric on $\text{SU}(2)$ with $\text{mult}(\lambda_1) = 4$.

4 Boundary remarks

- *Scale invariance.* The round metric of any radius r has eigenvalues $k(k+2)/r^2$ with the same multiplicities $(k+1)^2$, so no rescaling escapes the counterexample. More generally every bi-invariant metric on $\text{SU}(2)$ is isometric to a round $\mathbb{S}^3$.

- *Formalization.* The discrete skeleton is machine-checked in pure Lean 4 (no Mathlib, no axioms, no `sorry`; see `lean4/`): the eigenvalue and multiplicity tables, the dimension count $\#\text{Hom}_k - \#\text{Hom}_{k-2} = (k+1)^2$ for $k \leq 12$, the explicit $k = 1$ case, the linear independence of the coordinate functions via the identity evaluation matrix, and the rotation isometry (norm preservation and permutation of the axis points).
- *Generically Zoll metrics may well have simple spectrum* (as expected on S^3 by analogy with negative-curvature results); that is irrelevant to the refuted statement, which is a *universal upper bound* claim falsified by an explicit Zoll metric the conjecture itself points at.